

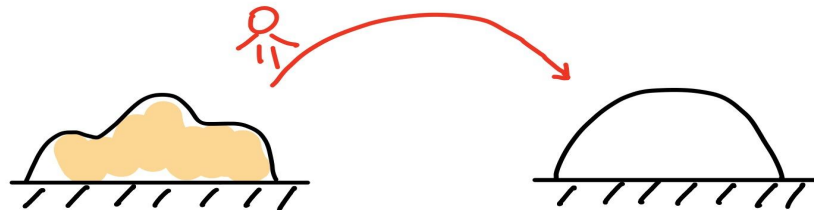


Fast Algorithms for Computational Optimal Transport and Wasserstein Barycenter

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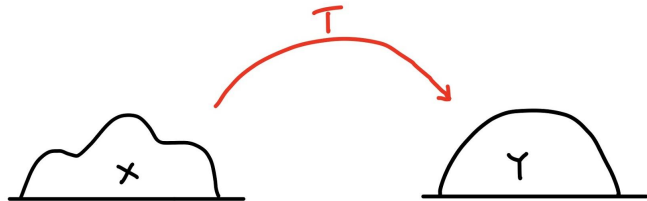
Optimal Transport

- Informally, consider a worker who need to move a pile of sand to a targeted sand castle



- The worker wants to minimize his total cost, e.g. distance or time spent carrying shovelfuls of sand
- What is the best way to achieve that?

Optimal Transport



- Mathematically, this problem can be casted as the comparison of two probability distributions.

*Given two probability distributions, what is the **optimal coupling** between them with **minimum cost**?*

- Optimal transport algorithms find the optimal way to “transport” from one measure to the other, with a minimal total cost
- The basic mathematical question has attracted much interest in theoretical research, with applications in graphics, computer vision, biology

Computational difficulty

Non-linearity in the constraint & **Scalability** to high dimension

→ Much prior work has been devoted to improving the **computational efficiency**

- Linear programming: Interior-point methods have been employed as a computational solver, with a practical complexity of $\tilde{O}(n^3)$ [PW09]
- Laplace linear system solver: improved the complexity of interior-point methods to $\tilde{O}(n^{2.5})$ [LS04] (dimension)
- Sinkhorn Algorithm: $\tilde{O}(\frac{n^2}{\epsilon})$ [S74, Kn08, Ka08, C13] (ϵ : approximation accuracy)
- **Recent accelerated primal-dual algorithms:** $\tilde{O}(\frac{n^{2.5}}{\epsilon})$ [D18, L19]

Contribution of this work

- **APDRCD algorithm:** Accelerated Primal-Dual Randomized Coordinate Descent
- **State-of-art complexity bound:** $\tilde{O}(\frac{n^{2.5}}{\epsilon})$, and more favorable empirical performance comparing to existing primal-dual algorithms
- We propose a **greedy** version of it, which further improves the empirical performance
- We demonstrate that these new algorithms can be **generalized to larger-scale** OT problems, e.g., approximating the Wasserstein barycenter

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Entropic-regularized OT

Formally, the optimization problem [C13] is :

$$\begin{aligned} \min_{X \in \mathbb{R}_+^{n \times n}} \quad & \langle C, X \rangle - \eta H(X) \\ \text{s.t.} \quad & X \mathbf{1} = r, \quad X^\top \mathbf{1} = l \end{aligned}$$

r, l : two known probability distributions in the probability simplex in Δ^n

C : the cost matrix, $C \in \mathbb{R}_+^{n \times n}$

X : transportation plan

$H(X)$: entropic regularization, $H(X) := - \sum_{i,j=1}^n X_{ij} \log(X_{ij})$

η : regularization parameter, $\eta > 0$

We aim to find an ϵ - approximate optimal: $\hat{X} : \langle C, \hat{X} \rangle \leq \langle C, X^* \rangle + \epsilon$

Dual formulation

- We can solve the entropic-regularized OT by solving its **dual** problem
- The Lagrangian:

$$\mathcal{L}(X, \alpha, \beta) := \langle C, X \rangle - \eta H(X) + \langle \alpha, r \rangle + \langle \beta, l \rangle - \langle \alpha, X \mathbf{1} \rangle - \langle \beta, X^\top \mathbf{1} \rangle$$

- Therefore, the **dual problem** is

$$\max_{\alpha, \beta \in \mathbb{R}^n} \min_{X \in \mathbb{R}^{n \times n}} \mathcal{L}(X, \alpha, \beta)$$

strictly convex in $X \rightarrow$ can solve exactly $X(\alpha, \beta)$

smooth function of α, β w.r.t. L2 norm [lemma 2.1]

APDRCD: Accelerated Primal-Dual Randomized Coordinate Descent

Algorithm 1: APDRCD ($C, \eta, A, b, \varepsilon'$)

1 **Input:** $\{\theta_i | \theta_0 = 1, \frac{1-\theta_{i+1}}{\theta_{i+1}^2} = \frac{1}{\theta_i^2}\}, C_0 = 1, \lambda^0 = z^0 =$
 $k = 0, L = \frac{4}{\eta}$

} Initialize with an auxiliary **decreasing** sequence θ_i

2 **while** $\|Ax^k - b\|_1 > \varepsilon'$ **do**

3 Set $y^k = (1 - \theta_k)\lambda^k + \theta_k z^k$

} Update X by a key **weighted average** of the dual variables, more recent ones have more weights

4 Compute $x^k = \frac{1}{C_k} \left(\sum_{j=0}^k \frac{x(y^j)}{\theta_j} \right)$

5 **Randomly sample one coordinate** i_k **where**
 $i_k \in \{1, 2, \dots, 2n\}$:

6 Update

$$\lambda_{i_k}^{k+1} = y_{i_k}^k - \frac{1}{L} \nabla_{i_k} \varphi(y^k) \quad (5)$$

} φ : dual objective function

7 Update

$$z_{i_k}^{k+1} = z_{i_k}^k - \frac{1}{2nL\theta_k} \nabla_{i_k} \varphi(y^k) \quad (6)$$

} Update the dual variables using an **accelerated randomized coordinate descent subroutine**

8 Update $k = k + 1$ and $C_k = C_k + \frac{1}{\theta_k}$

9 **end**

10 **Output:** X^k where $x^k = \text{vec}(X^k)$

➡ Output an ϵ - optimal solution

Complexity guarantee

Theorem

The APDRCD algorithm for approximating OT returns $\hat{X} \in \mathbb{R}^{n \times n}$ satisfying $\hat{X}\mathbf{1} = r, \hat{X}^T\mathbf{1} = l$ and $\mathbb{E}[\langle C, \hat{X} \rangle] - \langle C, X^* \rangle \leq \epsilon$ in a total number of

$$\mathcal{O}\left(\frac{n^{\frac{5}{2}} \sqrt{\|C\|_{\infty} \log(n)}}{\epsilon}\right)$$

arithmetic operations.

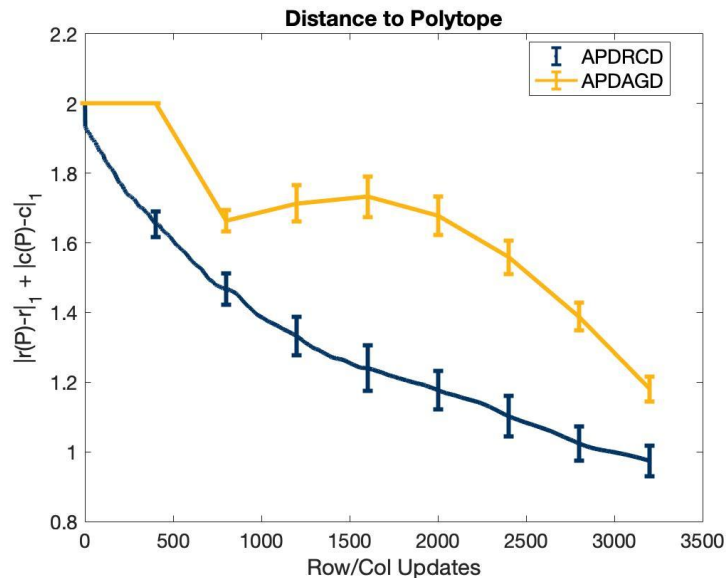
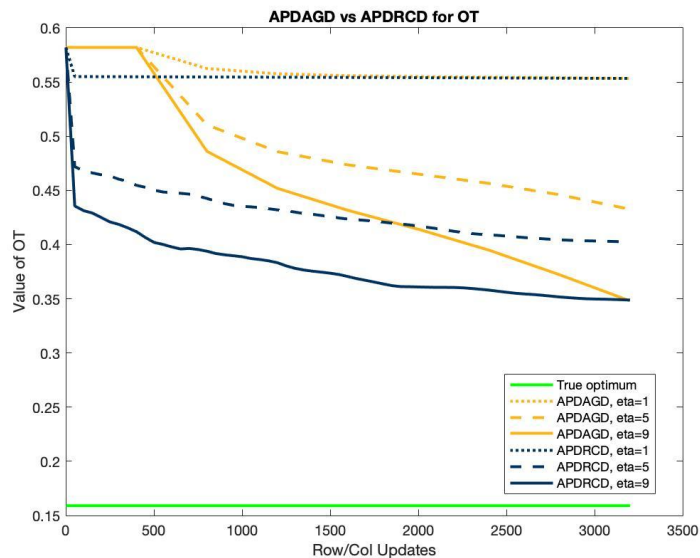
This complexity bound **matches with the state-of-art primal-dual algorithms**, with a **better dependence** on ϵ than the Sinkhorn algorithm ($\tilde{\mathcal{O}}(\frac{n^2}{\epsilon^2})$)

Proof idea: The proof first exploits the **smoothness** of the dual function, and obtains an **upper bound of the dual objective** (in expectation). Then such upper bound is translated to an **lower bound on the decrease** of the primal objective.

Experiments

- Compared with two other state-of-art primal dual algorithms for solving OT: APDAGD [D18] and APDAMD [L19]
- Datasets:
 - Synthetic Images [A17, L19]
 - MNIST [LeCun98]
 - CIFAR10 [A09]

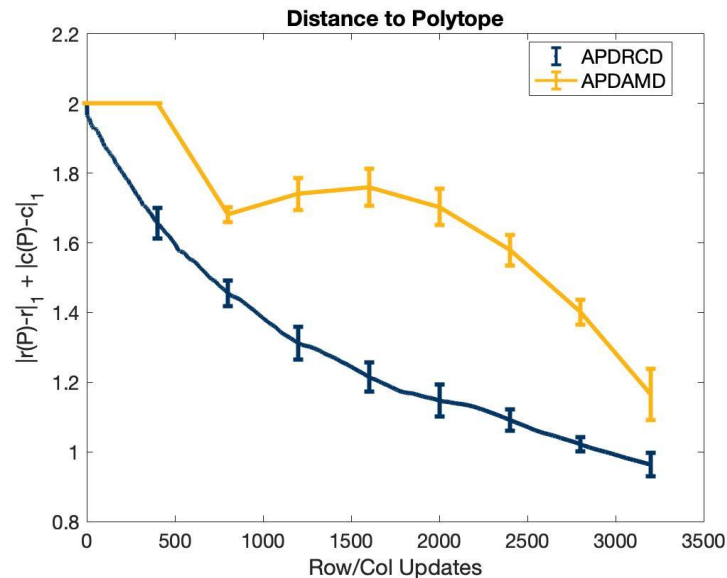
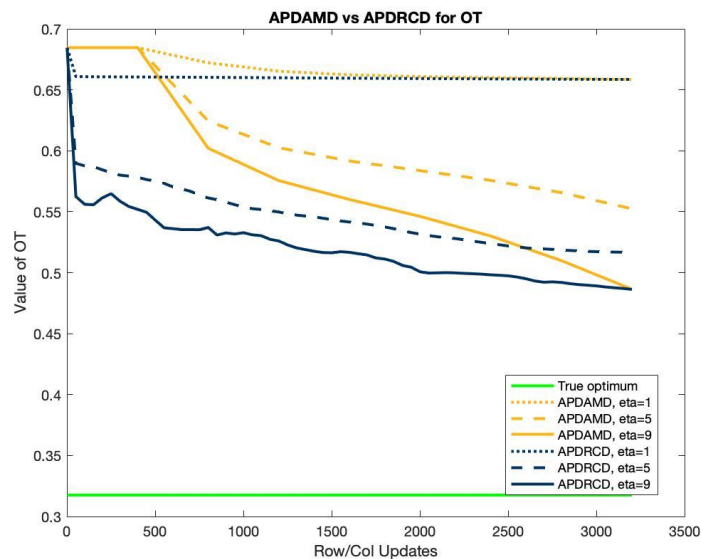
Experiments: synthetic images, compared with APDAGD [D18]



(left): Objective OT value vs Iterations. (Faster decrease is better)

(right): Violations to the marginal distributions vs Iterations. (Lower is better)

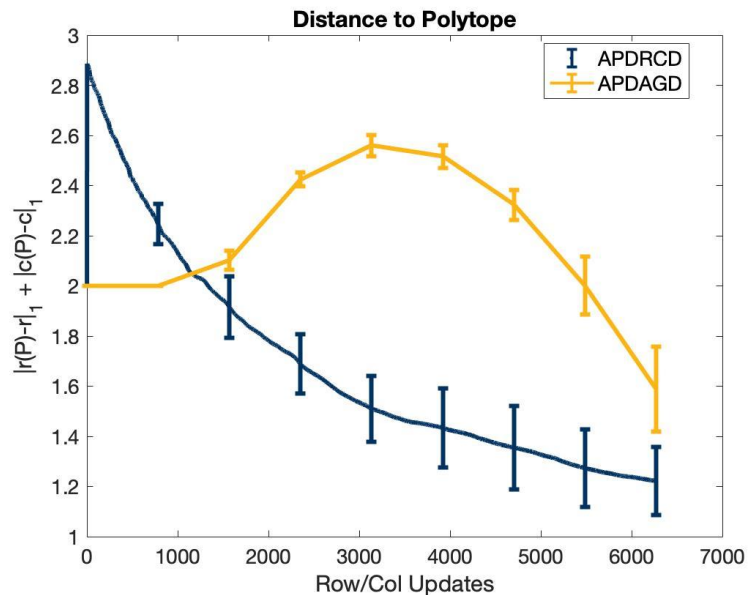
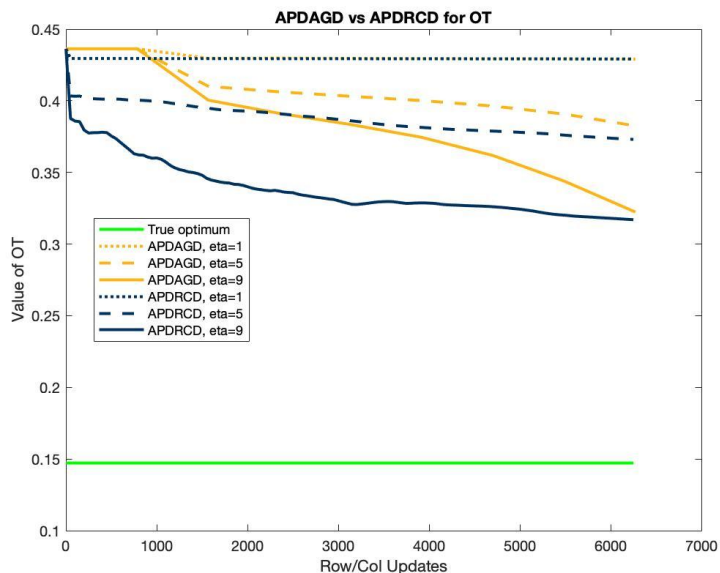
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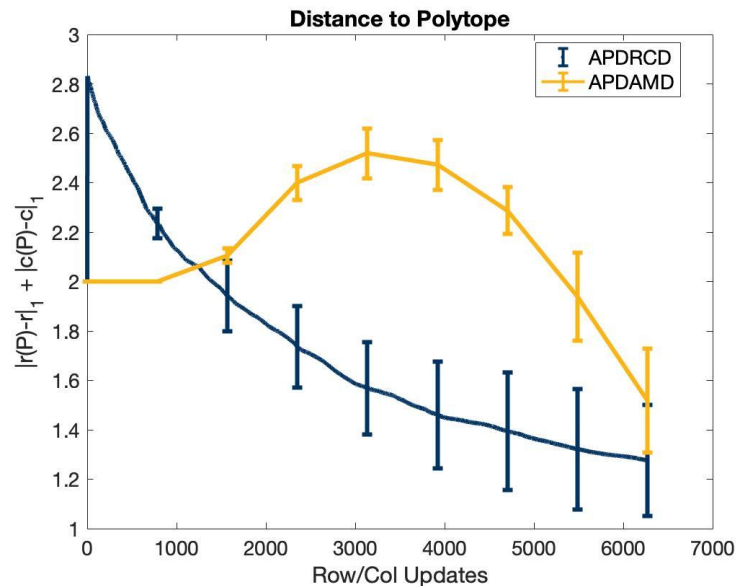
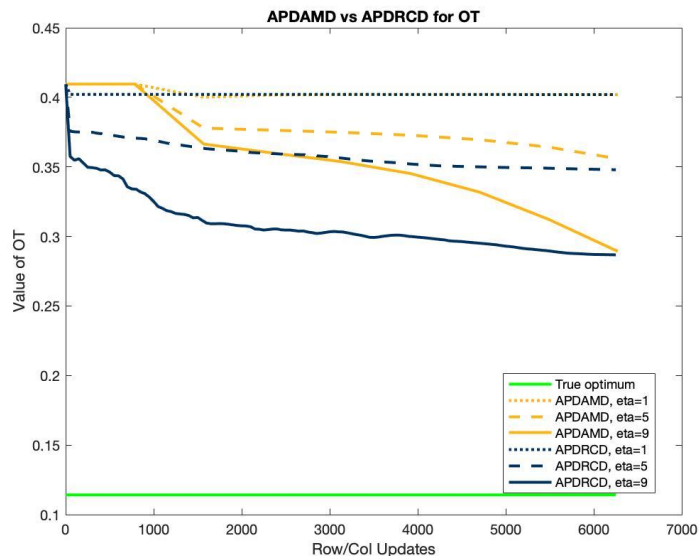
Experiments: MNIST, compared with APDAGD [D18]



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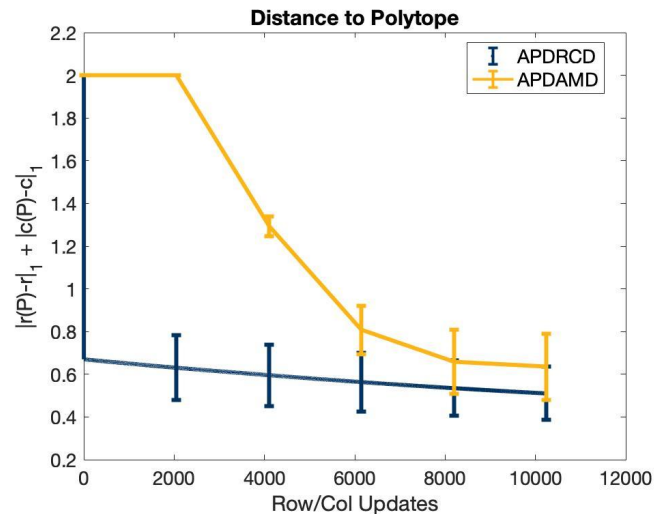
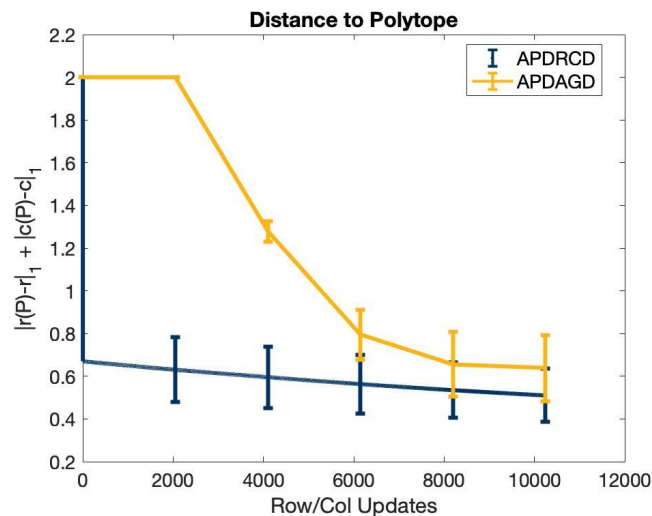
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Experiments: CIFAR10, compared with APDAGD [P18], APDAMD [L19]



(left): Objective OT value vs Iterations, compared with APDAGD [P18]. (Lower is better)
(right): Objective OT value vs Iterations, compared with APDAMD [L19].

More results

- We further propose a **greedy** version of APDRCD, which has the same algorithmic structure, but uses an accelerated greedy coordinate descent subroutine.

Further experiments show that this greedy version improves the empirical performance.

- We further demonstrate that these new algorithms can be **generalized to larger-scale** problems, e.g., computing the Wasserstein barycenter for multiple probability distributions.

Future work

- Though having matched complexity upper bounds, the primal-dual OT algorithms have different empirical performances. Further understanding on the selection of algorithms based on different real-world applications are definitely of interest.
- More empirical evaluations of these algorithms on larger scale problems, e.g. computing Wasserstein barycenters for multiple probability distributions.



Thank you!

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